

Bank Customer Churn Prediction & Business Analytics

Project Summary

This project analyses customer demographics, banking behaviour, and account characteristics to identify factors influencing customer churn. Using Python, machine learning, and Power BI, the project develops predictive models and interactive dashboards to support customer retention strategies and data-driven business decisions.

Key Business Insights

1. Customer Churn Overview

- The analysis identified a distinct segment of customers who exited the bank, highlighting the importance of proactive retention strategies.
- Understanding customer churn enables organizations to reduce customer attrition and improve long-term profitability.

2. Customer Demographics

- Customer churn varies across demographic groups such as **age, gender, and geography**.
- Segmenting customers by demographic characteristics helps identify high-risk customer groups and supports targeted engagement strategies.

3. Geographic Analysis

- Churn behavior differs across geographical regions, indicating that customer retention strategies may need to be tailored for different markets.
- Regional analysis helps organizations allocate resources more effectively and identify location-specific challenges.

4. Product Usage Analysis

- The number of products owned by a customer influences churn behavior.
- Customers with lower product engagement may require personalized offers and cross-selling initiatives to improve retention.

5. Active Member Analysis

- Customer activity levels are closely associated with retention.
- Monitoring active and inactive customers enables banks to identify disengaged customers and implement timely interventions.

6. Predictive Analytics

Two classification models were developed to predict customer churn:

- Logistic Regression
- Random Forest Classifier

The models were evaluated using:

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC

Performance evaluation demonstrates the effectiveness of machine learning in identifying customers who are likely to leave the bank.

Dashboard Capabilities

The interactive Power BI dashboard enables users to:

- Monitor overall customer churn.
- Analyze churn by geography.
- Compare churn across gender and age groups.
- Evaluate customer activity and product usage.
- View customer distribution through interactive visualizations.
- Filter insights dynamically using slicers.

Business Recommendations

Based on the analysis, the following recommendations are proposed:

- Develop targeted retention programs for customers identified as having a higher likelihood of churn.
- Increase customer engagement through personalized products and services.
- Monitor inactive customers regularly and initiate proactive communication.
- Strengthen cross-selling initiatives to improve product adoption and customer loyalty.
- Continuously monitor churn trends using interactive dashboards to support strategic decision-making.

Conclusion

This project demonstrates an end-to-end data analytics and machine learning workflow, including data preprocessing, exploratory data analysis, predictive modelling, model evaluation, and interactive dashboard development. The insights generated can help financial institutions understand customer behaviour, improve retention strategies, and support evidence-based business decisions.